

William O. Balmer

Curriculum Vitae

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Education

Johns Hopkins University, Baltimore, MD, USA Doctorate in Philosophy in Astrophysics	May 2023 – June 2026
Johns Hopkins University, Baltimore, MD, USA Masters in Physics	June 2021 – May 2023
Amherst College, Amherst, MA, USA BA <i>cum laude</i> in Astronomy; BA <i>cum laude</i> in Physics	August 2017 – May 2021

Positions

51 Pegasi b Fellow , Northwestern University	2026–
George E. Owen Fellow , Johns Hopkins University	2021–2024
Maryland Space Grant Observatory Fellow , Johns Hopkins University	2022–2023
Research Assistant , Follette Lab, Amherst College	2018–2021
Research Assistant , SIOS Lab, Cornell University	2020

Grants and Awards

Heising Simons Foundation 51 Pegasi b Fellow (2026–) **Awardee**, \$450,000
 James Webb Space Telescope Program GO-6915 (2025) **PI**, \$286,747
 James Webb Space Telescope Program GO-6905 (2025) **PI**, \$172,614
 James Webb Space Telescope Program DD-4558 (2024) **co-PI**, \$50,000
 James Webb Space Telescope Program GO-3337 (2024) **co-PI**, \$170,704
 Hubble Space Telescope Program GO-17122 (2024) **co-PI**, \$132,841
 NASA WIYN Data Award (2023) **PI**, \$6,000
 NASA WIYN Data Award (2022) **PI**, \$6,000
 Sigma Xi Scientific Honor Society, Full Member (2021)
 George E. Owen Fellowship (2021–2024) **Awardee**, \$18,000
 Amherst Memorial Fellowship (2021, 2022, 2023) **Awardee**
 Charles Hamilton Houston Award (2020) **Awardee**, \$4,500
 Gregory S. Call Student Researcher Award (2019) **Awardee**, \$3,500
 Sarles Fellow Award (2018) **Awardee**, \$3,500

Students Advised

Ash Messier (2025–present), Johns Hopkins University grad student.
 Evelyn Bruinsma (2025–present), Johns Hopkins University grad student.
 Gavin Wang (2023–present), Johns Hopkins University undergraduate. Goldwater Scholarship Winner.
 Klara Matuszewska (2025), STScI summer intern.
 Henry Dennen (2024), Johns Hopkins University summer intern.
 Matthew Prem (2022–2023), University of Maryland undergraduate.

Publications

First-author († indicates co-first authorship/equal contribution)

7. **Balmer, W. O.**, Pueyo, L., Messier, A., et al. (in press) *ApJL. Direct Images of CO₂ Absorption in the Atmosphere of a Super-Jupiter: Enhanced Metallicity Suggestive of Formation in a Disk*
6. †Bardalez Gagliuffi, D. C., †**Balmer, W. O.**, Pueyo, L., et al. (2025) *ApJL*, 988, L18. *JWST Coronagraphic Images of 14 Her c: A Cold Giant Planet in a Dynamically Hot Multiplanet System*
5. **Balmer, W. O.**, Kammerer, J., Pueyo, L., et al. (2025) *AJ*, 169, 209. *JWST-TST High Contrast: Living on the Wedge, or, NIRCам Bar Coronagraphy Reveals CO₂ in the HR 8799 and 51 Eri Exoplanets' Atmospheres*
4. **Balmer, W. O.**, Franson, K., Chomez, A., et al. (2025) *AJ*, 169, 30. *VLT/GRAVITY Observations of AF Lep b: Preference for Circular Orbits, Cloudy Atmospheres, and a Moderately Enhanced Metallicity*
3. **Balmer, W. O.**, Pueyo, L., Lacour, S., et al. (2024) *AJ*, 167, 64. *VLT/GRAVITY Provides Evidence the Young, Substellar Companion HD 136164 Ab Formed Like a “Failed Star”*
2. **Balmer, W. O.**, Pueyo, L., Stolker, T., et al. (2023) *ApJ*, 956, 99. *VLT/GRAVITY Observations and Characterization of the Brown Dwarf Companion HD 72946 B*
1. **Balmer, W. O.**, Follette, K. B., Close, L. M., et al. (2022) *AJ*, 164, 29. *Improved Orbital Constraints and H α Photometric Monitoring of the Directly Imaged Protoplanet Analog HD 142527 B*

Second-author (** indicates first author advised by W.O.B)

3. **Wang, G., **Balmer, W. O.**, Pueyo, L., et al. (2025) *AJ*, 169, 336. *A Revised Density Estimate for the Largest Known Exoplanet, HAT-P-67 b*
2. Franson, K., **Balmer, W. O.**, Bowler, B. P., et al. (2024) *ApJL*, 974, L11. *JWST/NIRCам 4–5 μ m Imaging of the Giant Planet AF Lep b*
1. Blunt, S., **Balmer, W. O.**, Wang, J. J., et al. (2023) *AJ*, 166, 257. *First VLT/GRAVITY Observations of HIP 65426 b: Evidence for a Low or Moderate Orbital Eccentricity*

Co-author

47. Denis, A., Vigan, A., Chauvin, G., et al. (2026) *A&A*, 707, L13. *Eccentric and cool: A high-spectral-resolution view of 51 Eri b with VLT/HiRISE*
46. Xuan, J. W., Ruffio, J.-B., Chachan, Y., et al. (2026) *ApJ*, 1000, 27. *The Compositions of the HR 8799 Planets Reflect Accretion of Both Solids and Metal-enriched Gas*
45. Ruffio, J.-B., Xuan, J. W., Chachan, Y., et al. (2026) *NatAs.tmp.*,. *Jupiter-like uniform metal enrichment in a system of multiple giant exoplanets*
44. Winterhalder, T. O., Mérand, A., Kammerer, J., et al. (2026) *A&A*, 705, A216. *Astrometric exomoon detection by means of optical interferometry*
43. Kral, Q., Wang, J., Kammerer, J., et al. (2026) *A&A*, 705, A217. *Exomoon search with VLT/GRAVITY around the substellar companion HD 206893 B*
42. Millar-Blanchaer, M. A., Choquet, É., Lawson, K., et al. (2025) *ApJ*, 994, 199. *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. VI. Evidence for Radially Evolving Icy Grains in the HD 141569A Disk via NIRCам Coronagraphic Imaging*
41. Palatnick, S., Millar-Blanchaer, M. A., Zhang, J., et al. (2025) *ApJ*, 995, 149. *Discovery of a Debris Disk around TWA 20*
40. Houllé, M., Millour, F., Berio, P., et al. (2025) *A&A*, 704, A182. *The mid-infrared spectrum of β Pictoris b: First VLT/MATISSE interferometric observations of an exoplanet*
39. Kammerer, J., Winterhalder, T. O., Lacour, S., et al. (2025) *A&A*, 704, A318. *The ExoGRAVITY survey: A K-band spectral library of giant exoplanet and brown dwarf companions*
38. Mâlin, M., Ward-Duong, K., Grant, S. L., et al. (2025) *A&A*, 704, A181. *JWST-TST High Contrast: Medium-resolution spectroscopy reveals a carbon-rich circumplanetary disk around the young accreting exoplanet Delorme 1 AB b*
37. Venkatesan, V., Blunt, S., Wang, J. J., et al. (2025) *ApJ*, 993, 69. *Constraints on the Orbit of the Young Substellar Companion GQ Lup B from High-resolution Spectroscopy and VLT/GRAVITY Astrometry*

36. Manjavacas, E., Gennaro, M., Betti, S., et al. (2025) *ApJ*, 994, 64. *A Near-infrared Spectral Library of Very Young Brown Dwarfs and Planetary Mass Objects in the Orion Nebula Cluster*
35. Roberts, J., Thompson, W., Wang, J. J., et al. (2025) *AJ*, 170, 273. *New Orbital Constraints for YSES 1 b and HR 2562 B from High-Precision Astrometry and Planetary Radial Velocities*
34. Bogat, E., Schlieder, J. E., Lawson, K. D., et al. (2025) *AJ*, 170, 225. *Probing the Outskirts of M Dwarf Planetary Systems with a Cycle 1 JWST NIRCам Coronagraphy Survey*
33. Kenworthy, M., Stolker, T., Kammerer, J., et al. (2025) *A&A*, 701, A104. *YSES 2b is a background star: Differential astrometric M-dwarf measurements in time*
32. Beichman, C., Sanghi, A., Mawet, D., et al. (2025) *ApJL*, 989, L22. *Worlds Next Door: A Candidate Giant Planet Imaged in the Habitable Zone of α Centauri A. I. Observations, Orbital and Physical Properties, and Exozodi Upper Limits*
31. Winterhalder, T. O., Kammerer, J., Lacour, S., et al. (2025) *A&A*, 700, A4. *Orbit and atmosphere of HIP 99770 b through the eyes of VLTI/GRAVITY*
30. Stolker, T., Samland, M., Waters, L. B. F. M., et al. (2025) *A&A*, 700, A21. *Direct imaging discovery of a young giant planet orbiting on Solar System scales*
29. Sanghi, A., Beichman, C., Mawet, D., et al. (2025) *ApJL*, 989, L23. *Worlds Next Door: A Candidate Giant Planet Imaged in the Habitable Zone of α Centauri A. II. Binary Star Modeling, Planet and Exozodi Search, and Sensitivity Analysis*
28. Crotts, K. A., Carter, A. L., Lawson, K., et al. (2025) *ApJL*, 987, L41. *Follow-up Exploration of the TWA 7 Planet-Disk System with JWST NIRCам*
27. Hoch, K. K. W., Rowland, M., Petrus, S., et al. (2025) *Natur*, 643, 938-942. *Silicate clouds and a circum-planetary disk in the YSES-1 exoplanet system*
26. Hayoz, J., Bonse, M. J., Dannert, F., et al. (2025) *A&A*, 698, A87. *High-contrast spectroscopy with the new VLT/ERIS instrument: Molecular maps and radial velocity of the gas giant AF Lep b*
25. Trevascus, D., Blunt, S., Christiaens, V., et al. (2025) *A&A*, 698, A19. *Differentiating formation models with new dynamical masses for the PDS 70 protoplanets*
24. Sanghi, A., Beichman, C., Mawet, D., et al. (2025) *RNAAS*, 9, 119. *A Preliminary Search for Planets and Exozodiacal Emission Around α Centauri A with JWST/MIRI*
23. Denis, A., Vigan, A., Costes, J., et al. (2025) *A&A*, 696, A6. *Characterization of AF Lep b at high spectral resolution with VLT/HIRISE*
22. Ray, S., Sallum, S., Hinkley, S., et al. (2025) *ApJL*, 983, L25. *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. III. Aperture Masking Interferometric Observations of the Star HIP 65426 at 3.8 μ m*
21. Chai, Y., Chen, C. H., Worthen, K., et al. (2024) *ApJ*, 976, 167. *A JWST MIRI MRS View of the η Tel Debris Disk and Its Brown Dwarf Companion*
20. Maire, A.-L., Leclerc, A., Balmer, W. O., et al. (2024) *A&A*, 691, A263. *Direct imaging and dynamical mass of a benchmark T-type brown dwarf companion to HD 167665*
19. Xuan, J. W., Mérand, A., Thompson, W., et al. (2024) *Natur*, 634, 1070-1074. *The cool brown dwarf Gliese 229 B is a close binary*
18. Hoch, K. K. W., Theissen, C. A., Barman, T. S., et al. (2024) *AJ*, 168, 187. *JWST-TST High Contrast: Spectroscopic Characterization of the Benchmark Brown Dwarf HD 19467 B with the NIRSpec Integral Field Spectrograph*
17. Blunt, S., Wang, J., Hirsch, L., et al. (2024) *JOSS*, 9, 6756. *orbitize! v3: Orbit fitting for the High-contrast Imaging Community*
16. Ruffio, J.-B., Perrin, M. D., Hoch, K. K. W., et al. (2024) *AJ*, 168, 73. *JWST-TST High Contrast: Achieving Direct Spectroscopy of Faint Substellar Companions Next to Bright Stars with the NIRSpec Integral Field Unit*
15. Kammerer, J., Lawson, K., Perrin, M. D., et al. (2024) *AJ*, 168, 51. *JWST-TST High Contrast: JWST/NIRCам Observations of the Young Giant Planet β Pic b*
14. Winterhalder, T. O., Lacour, S., Mérand, A., et al. (2024) *A&A*, 688, A44. *Combining Gaia and GRAVITY: Characterising five new directly detected substellar companions*

13. Nowak, M., Lacour, S., Abuter, R., et al. (2024) *A&A*, 687, A248. *Catalogue of dual-field interferometric binary calibrators*
12. Nasedkin, E., Mollière, P., Lacour, S., et al. (2024) *A&A*, 687, A298. *Four-of-a-kind? Comprehensive atmospheric characterisation of the HR 8799 planets with VLTI/GRAVITY*
11. Pourré, N., Winterhalder, T. O., Le Bouquin, J.-B., et al. (2024) *A&A*, 686, A258. *High contrast at short separation with VLTI/GRAVITY: Bringing Gaia companions to light*
10. Petrus, S., Whiteford, N., Patapis, P., et al. (2024) *ApJL*, 966, L11. *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. V. Do Self-consistent Atmospheric Models Represent JWST Spectra? A Showcase with VHS 1256–1257 b*
9. Worthen, K., Chen, C. H., Law, D. R., et al. (2024) *ApJ*, 964, 168. *MIRI MRS Observations of β Pictoris. I. The Inner Dust, the Planet, and the Gas*
8. Sallum, S., Ray, S., Kammerer, J., et al. (2024) *ApJL*, 963, L2. *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems. IV. NIRISS Aperture Masking Interferometry Performance and Lessons Learned*
7. Grant, D., Lewis, N. K., Wakeford, H. R., et al. (2023) *ApJL*, 956, L32. *JWST-TST DREAMS: Quartz Clouds in the Atmosphere of WASP-17b*
6. Carter, A. L., Hinkley, S., Kammerer, J., et al. (2023) *ApJL*, 951, L20. *The JWST Early Release Science Program for Direct Observations of Exoplanetary Systems I: High-contrast Imaging of the Exoplanet HIP 65426 b from 2 to 16 μ m*
5. Follette, K. B., Close, L. M., Males, J. R., et al. (2023) *AJ*, 165, 225. *The Giant Accreting Protoplanet Survey (GAPlanetS)-Results from a 6 yr Campaign to Image Accreting Protoplanets*
4. Hinkley, S., Lacour, S., Marleau, G.-D., et al. (2023) *A&A*, 671, L5. *Direct discovery of the inner exoplanet in the HD 206893 system. Evidence for deuterium burning in a planetary-mass companion*
3. Miles, B. E., Biller, B. A., Patapis, P., et al. (2023) *ApJL*, 946, L6. *The JWST Early-release Science Program for Direct Observations of Exoplanetary Systems II: A 1 to 20 μ m Spectrum of the Planetary-mass Companion VHS 1256-1257 b*
2. Adams Redai, J. I., Follette, K. B., Wang, J., et al. (2023) *AJ*, 165, 57. *The Giant Accreting Protoplanet Survey (GAPlanetS): Optimization Techniques for Robust Detections of Protoplanets*
1. Betti, S. K., Follette, K. B., Ward-Duong, K., et al. (2022) *ApJL*, 935, L18. *Near-infrared Accretion Signatures from the Circumbinary Planetary-mass Companion Delorme 1 (AB)b*

Selected Press

NASA's Webb Redefines Dividing Line Between Planets, Stars (Balmer et al. 2026):

[NASA Press Release](#),

Frigid Exoplanet in Strange Orbit Imaged by NASA's Webb (Bardalez-Gagliuffi & Balmer et al. 2025):

[NASA Press Release](#), [Forbes](#), [Space.com](#)

Scientists used JWST instruments 'wrong' on purpose to capture direct images of exoplanets

(Balmer et al. 2025b): [Space.com](#)

NASA's Webb Images Young, Giant Exoplanets, Detects Carbon Dioxide (Balmer et al. 2025b):

[JHU and NASA Press Release](#), [CBS](#), [AFP](#), [El Mundo](#), [Futurism.com](#), [Mashable.com](#), [BBC Sky at Night](#)

UT Astronomers Race To Capture Image of Exoplanet Near Star (Franson et al. 2024):

[McDonald Observatory Press Release](#)

Newfound alien planet has nuclear fusion going in its core (Hinkley et al. 2023)

[Space.com](#)

Observing Programs

JWST GO-6915, Cycle 4 **PI**, 48 hours

JWST GO-6905, Cycle 4 **PI**, 24 hours

MINERVA-A NN-EXPLORE, 2025A **Co-PI**, 23 hours

VLTI/GRAVITY, P114 **PI**, 7 hours

JWST DD-4558, Cycle 2 **Co-PI**, 6.4 hours

VLTI/GRAVITY, P112 **PI**, 9 hours

JWST GO-3337, Cycle 2 **Co-PI**, 7.6 hours
 VLT/GRAVITY, P111-116 **PI**, 15 hours
 HST GO-17122, Cycle 30 **Co-PI**, 9 orbits
 VLT/GRAVITY, P109 **PI**, 3 hours
 WIYN 3.5m NN-EXPLORE, 2022A **PI**, 2.4 nights
 SOAR 4.1m NOIRLAB, 2022A **PI**, 2 nights
 ARC 3.5m APO, 2021.4 **PI**, 24 hours
 200+ hours of JWST observing time as **Co-I**
 100+ hours of VLT(I) observing time as **Co-I**

Selected Presentations

Contributed

- JWST Coronagraphic Imaging of Eccentric, Cold Giants (talk)*
 30th Anniversary of 51 Peg b: Cool Giant Planets and Their Systems, Observatoire de Haute-Provence, FR October 2025
- JWST Coronagraphic Images of 14 Her c: a Cold Giant Planet ... (talk)*
 American Astronomical Society 246th Meeting, Anchorage, AK June 2025
- The Bleeding Edges of Direct Imaging with JWST (talk)*
 NASA ExoPAG 31, National Harbor, MD January 2025
- Long baseline optical interferometry of exoplanets and brown dwarfs (talk)*
 Chesapeake Bay Area Exoplanet Meeting #11, Baltimore, MD May 2024
- Direct Detection and Characterization of Ice-line Giants with Optical Interferometry (talk)*
 Pathways to Characterizing Non-Transiting Planets, Greenbelt, MD April 2024
- The Unexpected Detection of HR8799e with NIRCcam Coronagraphy ... (talk)*
 Planetary Systems and the Origins of Life in the Era of JWST, STScI Spring Symposium, Baltimore, MD May 2023
- Unprecedented precision: using the VLTI jointly with Gaia ... (talk)*
 Cool Stars 21, Toulouse, FR August 2022
- Improved Orbital Constraints and H α Photometric Monitoring ... (poster)*
 Cool Stars 21, Toulouse, FR August 2022

Invited Seminars

- Directly Imaging Giant Planets with JWST*
 Observation Seminar, Northwestern University, Evanston, IL November 2025
- Ibid.*
 Star and Planet Formation Seminar, University of Michigan, Ann Arbor, MI September 2025
- Ibid.*
 Planetary Astronomy Lunch Seminar, University of Maryland, College Park, MD September 2025
- Ibid.*
 Institute for Astronomy Star and Planet Seminar, Honolulu, HI September 2025
- Living on the Wedge: New insights from Bar Coronagraphy*
 NIRCcam Builders Meeting, Biosphere 2, Tucson, AZ March 2025
- Ibid.*
 Max Planck Institute for Astronomy ExoCoffee, Heidelberg, DE January 2025
- Direct detection of a cold giant planet in a dynamically hot system*
 JHU ExoJamboree, Baltimore, MD November 2024
- Ibid.*
 Carnegie Earth and Planetary Laboratory Astronomy Seminar, Washington DC October 2024
- The formation of AF Leporis b*
 Protoplanets Group, Observatoire de la Côte d'Azur, Nice, FR June 2024
- Ibid.*
 European Southern Observatory Star and Planet Formation Seminar, Garching, DE November 2023
- The 4-5 micron color of HR 8799*
 European Southern Observatory Stellar Coffee and Planetary Tea, Garching, DE November 2023
- Ibid.*
 ExoGRAVITY Collaboration Workshop, Heidelberg, DE November 2023

Ibid.

petitRADTRANS Collaboration Workshop, Heidelberg, DE

November 2023

Benchmark Brown Dwarfs at Extreme Precision

American Museum of Natural History Astronomy Colloquium, New York City, NY

February 2023

Ibid.

Space Telescope Science Institute HotSci Series, Baltimore, MD

August 2022

Service and Outreach

Referee/review panel service:

- Reviewer, Consolidator Grant, European Research Council (2025)
- Executive Secretary, NASA XRP Panel (2024)
- Referee for Astronomy & Astrophysics (2025-)
- Referee for The Astrophysical Journal (2025-)
- Referee for The Astronomical Journal (2022-)

Organization service:

- JHU ExoJamboree SOC (2025)
- JHU Exoplanet No-PhDs Journal Club Lead Organizer (2022-)

Outreach activities:

- Observatory Open Houses, K-12 Tours, Maryland Space Grant Observatory (2022-)
- Pen Pal, Letters to a Pre-Scientist (2024-)
- Editor, *Astrobites* (2023-2024)
- Author, *Astrobites* (2021-2023)
- Astronomy Editor, *The Amherst STEM Network* (2019-2021)
- Observatory Operator, Amherst College Observatory (2021)

Public Lectures:

- Invited talk, North County High School (2023)
- Invited talk, Howard Astronomical League (2022)
- Invited talk, Balticon 56 (2022)
- Invited Talk, UMass Amherst Astronomy Club (2021)